

Técnicas e Algoritmos em Ciência de Dados

A1 Exam

Answer all the questions.

Time limit: 1.5 hours.

Late submissions will NOT be accepted.

Instructions

Identifier

Please choose a random number of 6 digits and write it at the beginning of your submission. Make sure that you keep a copy of that number (please avoid trivial numbers, such as 000000 or 123456 – thank you).

Submission

Submit only one file through ECLASS.

The name of the file will be the random number that identifies you (for example 568423.pdf)

If you have issues, at the very last minute, email your A1 exam as an attachment to alberto.paccanaro@fgv.br with the subject “URGENT – EXAM A1 SUBMISSION”. In the body of the message, explain the reason for not submitting through ECLASS.

All the work you submit should be solely your own work. Your submissions will be checked for this.

IMPORTANT: throughout the exam, keep your explanations short and to the point.

Question 1 -- value: 30%

You have just developed a binary classifier, to classify datapoints into two classes: C_1 (or Positive) and C_0 (or Negative). The confusion matrix summarizing the performance of your system on a test set is given below.

	Predicted	
Actual	TN 9490	FP 500
	FN 0	TP 10

Using Bayes theorem, estimate the probability of being in C_1 for a point that was predicted to be in C_1 by your binary classifier (exact calculations are not necessary, but you need to show the steps involved in applying Bayes theorem to this problem).

You can assume that the training test was “fair”, so that you can estimate probabilities from the values in the table above.

Question 2 – value: 30%

Figure 1 shows a 2-dimensional dataset, with points belonging to 3 different classes (C_0 , C_1 , C_2) represented by different colours (points from C_0 are red; points from C_1 are blue; points from C_2 are green). Note that:

- Each of the three classes has the same number of points.
- Each blob of points is Gaussian distributed.
- The different blobs of points belonging to the same class have the same number of points.

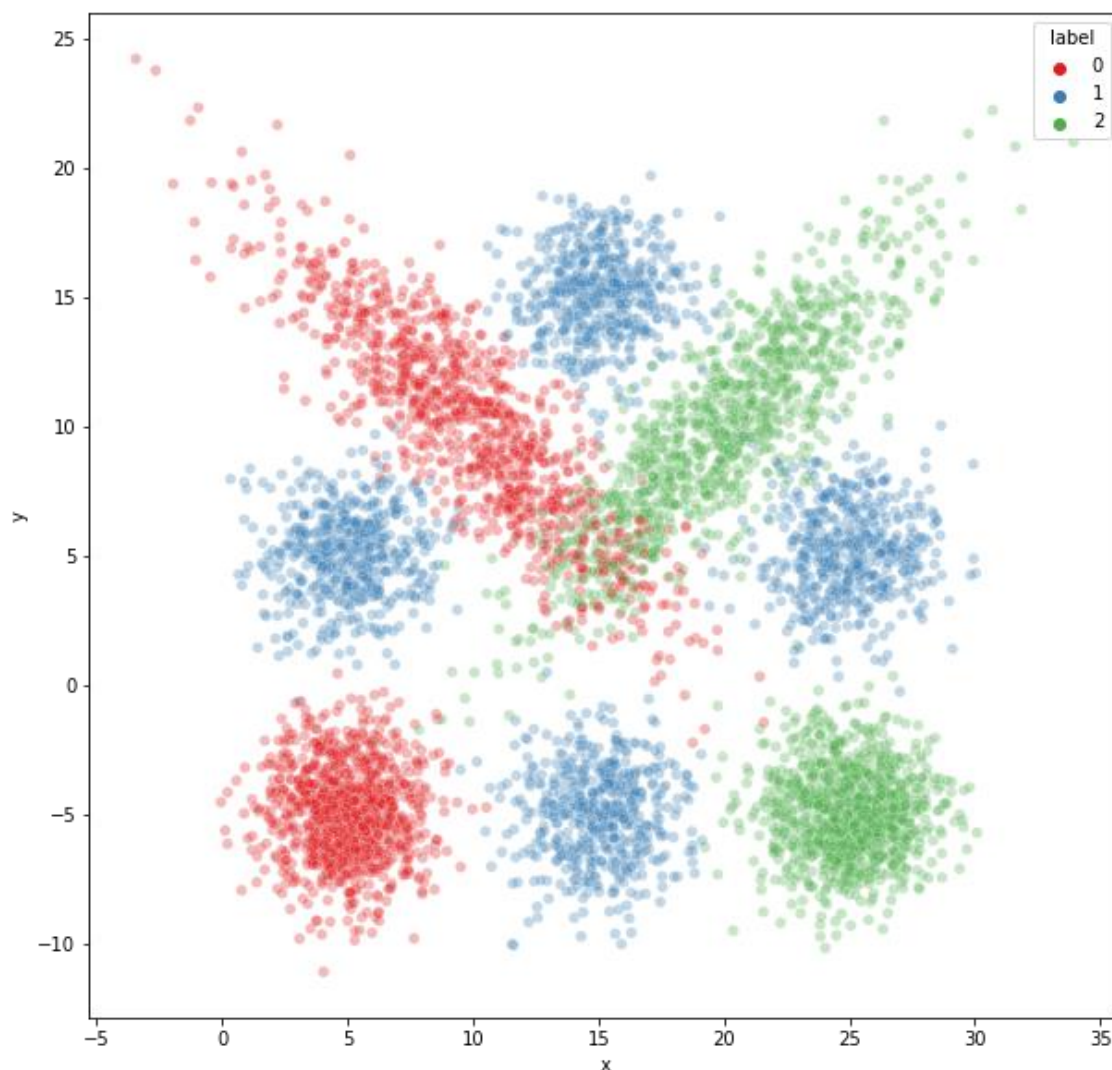


Figure 1

Consider the following classifiers:

- a) Naïve Bayes
- b) K-nearest neighbour.
- c) A very simple model that classifies datapoints using only the class priors
- d) A Bayesian classifier that fits a single gaussian for each class.
- e) A Bayesian classifier based on the Mixture of Gaussians model

Considering *accuracy* as the measure of performance, answer the following points:

1. For each of the classifiers, discuss why you think it should work well, or not, for the task of classifying the datapoints.
2. What *accuracy* would you expect for classifier c)
3. Provide a rough estimate of the position of the mean of the model of each class when using classifier d). Note that no calculation is necessary, just a rough estimate based on the figure will be sufficient.
4. State which model you think should work best. Justify your answer.

Question 3 – value: 20%

You are given the following 4 datapoints belonging to class C_0 : (1,5), (3,5), (3,7), (1,7), and the following 4 datapoints belonging to class C_1 : (0,2), (3,1), (6,2), (3,3). These are represented in Figure 2 as blue and red dots, respectively.

You need to classify point (3,4), represented as a green cross in Figure 2.

Assume the following:

- each of the classes are Gaussian distributed. That means that the points in C_0 are drawn from a single Gaussian distribution, and the points from C_1 are drawn from a single Gaussian distribution.
- the Gaussians have the same covariance matrix
- the Gaussians are spherical – that is, the elements of the off-diagonal elements of the covariance matrix are 0 and the covariance matrix for both Gaussians is: $\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$ with $a > 0$.

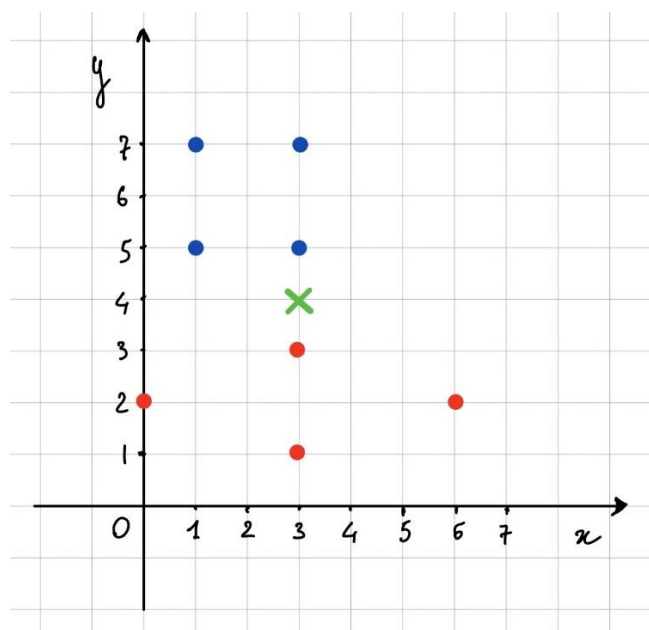


Figure 2

1. Classify the point (3,4) using a Bayesian classifier that models each of the classes as single Gaussians. Explain each step involved in building your Bayesian classifier and describe how you use it to classify the datapoint. You will need to provide explicitly the maximum likelihood estimates of the mean of your Gaussian models and the priors.
2. Classify the point (3,4) using K-nearest-neighbour with $K = 2$
3. Are the results from point 1) and 2) the same? In case they are different, discuss which solution you would think is more correct.

Tip: The calculations in this exercise should be trivial. Most answers can be inferred by considerations based on the geometry of the points in Figure 2.

Question 4 – value: 20%

Discuss the relationship between:

- Dimensionality of the dataset (that is, the dimensionality of the datapoints in your problem)
- Size of the dataset (that is, the number of datapoints that you have available)
- Complexity of the machine learning model

when learning from a dataset.

– END –